

Physical Promotion Attempt From Finite-Capacity Causal Networks

Abraham Greenman

2026

Abstract

This paper closes Paper 17 in the finite-capacity causal-network theorem ladder. Starting from the closed Paper 16 conditional external review and reproduction certificates theorem, it constructs a finite, network-native physical-promotion attempt interface. The construction is conditional and non-promoting: it defines attempt records, eligibility references, evidence bundle references, review certificate references, decision rows, objection rows, residual-risk rows, stability rows, audit rows, rollback rows, and a final conditional certificate. It does not prove physical promotion attempt success, physical promotion, physical validation, empirical adequacy, review acceptance, reproduction success, benchmark success, prediction success, falsification success, physical nature realization, or a unified field theory.

1 Theorem Status

The Paper 17 theorem is closed as a conditional finite interface theorem. The formal Lean endpoint is:

```
paper17_ppa008_final_conditional_certificate_closes  
physical_promotion_attempt_theorem
```

The closing contract is:

```
Paper17PhysicalPromotionAttemptTheoremContract.closed
```

The Rust audit surface is in `rust/cclab_accel/src/lib.rs`, and the regression tests are in `rust/cclab_accel/tests/physical_promotion_attempt.rs`. The required local checks are:

```
make test-fast  
make lean-build
```

2 Upstream Assumption

Paper 17 starts after the frozen Paper 16 result:

- Paper 16 commit: `8bb266493b1c53c776b2d39a47140794edc8935f`.
- Paper 16 formal endpoint: `Paper16ExternalReviewReproductionCertificatesTheoremContract.closed`.
- Paper 16 canonical proof: `paper16_errc008_final_conditional_certificate_closes_external_review_reproduction_certificates_theorem`.

The upstream result is consumed only as a conditional review/reproduction certificate package. It is not reinterpreted as review acceptance, reproduction success, external validation success, physical promotion, or physical nature realization.

3 Main Theorem

Theorem. Starting from the closed Paper 16 conditional external review and reproduction certificates theorem, there is a finite, network-native physical-promotion attempt interface preserving finite capacity, locality, bounded transfer, Paper 16 certificate compatibility, stability, auditability, rollback behavior, and no-hidden-import boundaries. The interface closes only as a conditional attempt-interface theorem and does not close any physical-promotion, physical-validation, success, physical nature, or unified-field claim.

4 Construction

The construction is a sequence of eight closed rungs.

4.1 PPA-001: Upstream Binding

PPA-001 binds Papers 1–16 through frozen upstream commits and the Paper 16 final conditional certificate. The rung preserves finite capacity, locality, bounded transfer, and the claim boundary. No physical promotion, validation, empirical adequacy, review acceptance, reproduction success, physical nature, or unified-field claim is imported.

4.2 PPA-002: Finite Attempt Records

PPA-002 defines the physical-promotion attempt record as a bounded, auditable interface row. Each row contains finite labels for:

- attempt identifier,
- eligibility reference,
- evidence-bundle reference,
- review-certificate reference,
- decision reference,
- objection reference,
- residual-risk reference,
- audit-status descriptor.

The row is an attempt record only. It does not assert attempt success, promotion, validation, empirical adequacy, acceptance, reproduction success, benchmark success, prediction success, falsification success, simulation-only promotion, fit-only calibration, physical nature, or unified-field promotion.

4.3 PPA-003: Eligibility, Evidence, and Review Descriptors

PPA-003 refines the attempt row with finite eligibility, evidence-bundle, and review-reference descriptors. Eligibility descriptors are not eligibility success findings. Evidence descriptors are not evidence sufficiency claims. Review references are not review acceptance claims. Paper 16 certificate references remain reference-only.

4.4 PPA-004: Decision, Objection, and Residual-Risk Descriptors

PPA-004 defines finite decision, objection, and residual-risk descriptors. Decision descriptors are not promotion decisions. Objection descriptors are not objection adjudications or review outcomes. Residual-risk descriptors are not risk discharge claims. These descriptors are bounded, auditable, local to the attempt interface, and rollback-compatible.

4.5 PPA-005: Paper 16 Certificate Compatibility

PPA-005 defines compatibility rows for the frozen Paper 16 endpoint and final certificate. The compatibility rows record finite references to review and reproduction certificates without asserting review acceptance, reproduction success, validation success, benchmark success, prediction success, or falsification success.

4.6 PPA-006: Stability, Audit, and Rollback

PPA-006 defines finite stability, audit, and rollback descriptors. Stability is interface stability only, not empirical stability. Audit descriptors are finite and replayable from attempt rows. Rollback is local bookkeeping for the attempt interface and is not falsification success, review failure, promotion failure, validation failure, or empirical inadequacy.

4.7 PPA-007: No-Hidden-Import Audit

PPA-007 closes a fail-closed audit for hidden imports. The audit rejects hidden physical-promotion attempt success, physical promotion, physical validation, empirical adequacy, review acceptance, reproduction success, benchmark success, prediction success, falsification success, simulation-only promotion, fit-only calibration, physical nature realization, and unified-field-theory imports. Closing this audit establishes absence of those hidden imports only; it does not promote or validate the attempt.

4.8 PPA-008: Final Conditional Certificate

PPA-008 assembles the final certificate. It consumes the closed contracts for PPA-001 through PPA-007 and closes `Paper17PhysicalPromotionAttemptTheoremContract.closed`. The closing certificate is conditional and interface-level. It preserves finite capacity, locality, bounded transfer, the hidden-import audit, and every non-promotion boundary.

5 Proof

The proof is by finite contract assembly. For each rung, the Lean development defines a contract whose `closed` predicate is a finite conjunction of its required fields. A canonical contract instantiates each field with the corresponding boundedness, compatibility, audit, or no-claim witness. Lean then proves each canonical rung closed by unfolding the contract and simplifying.

The final theorem contract contains the closed status of PPA-001 through PPA-008 plus the non-promotion boundary fields. The theorem:

```
paper17_ppa008_final_conditional_certificate_closes
physical_promotion_attempt_theorem
```

unfolds the final contract, unfolds the canonical rung witnesses, and closes the conjunction. Therefore the Paper 17 theorem is closed exactly as a conditional finite physical-promotion attempt interface theorem.

6 Runtime Audit

The Rust guard mirrors the formal ladder. Each rung has a canonical runtime record and a `closes_ppa00n` predicate. The final runtime certificate requires:

- all descriptors are bounded,
- the Paper 16 endpoint and final certificate match the frozen references,
- every compatibility row is reference-only,
- rollback is local bookkeeping,

- the hidden-import audit rejects all forbidden claims,
- all physical-promotion and success claims remain false.

The final runtime certificate closes the Paper 17 theorem contract only when all rung predicates close and the non-promotion boundary remains false for all forbidden claim fields.

7 Claim Boundary

This theorem does not prove any of the following:

- physical promotion attempt success,
- physical promotion,
- physical validation,
- empirical adequacy,
- review acceptance,
- reproduction success,
- benchmark success,
- prediction success,
- falsification success,
- observed particle catalog recovery,
- physical Standard Model recovery,
- physical particle excitation recovery,
- physical matter-field recovery,
- physical gauge-field recovery,
- physical quantum dynamics,
- continuum quantum field theory,
- simulation-only promotion,
- fit-only calibration,
- physical nature realization,
- unified field theory.

8 Reproducibility

The theorem can be checked locally from the repository root:

```
make test-fast
make lean-build
```

The Lean proof is located at `GPD/formal/FiniteCapacity/PhysicalPromotionAttempt.lean`. The Rust guard is located at `rust/cclab_accel/src/lib.rs`. The regression tests are located at `rust/cclab_accel/tests/physical_promotion_attempt.rs`.

9 Conclusion

Paper 17 closes the finite physical-promotion attempt interface theorem. It provides a bounded, auditable, rollback-compatible, certificate-compatible attempt structure downstream of Paper 16. The closure is intentionally narrow: it closes the existence and compatibility of the finite attempt interface, not promotion success, validation success, empirical adequacy, physical nature realization, or unified-field promotion.